

presidioAI

mapping out the law

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Motivation

People have legal problems, and they don't know where to start.

The first step to tackle any legal problem is to find past legal cases that were similar to yours.

That way, you can use it as a reference to understand your own situation, evaluate what you may need to prove to have success, or use that analogical case as evidence while you plead your case.

Legal argument is based on citing precedent, so it is important that people can accurately find and analogize their situation to another existing case. ChatGPT doesn't necessarily provide perfect answers since it is guessing based on token generation, and it doesn't provide transparency for users about how it is proposing cases. We hypothesize that building a specialized model to assess case similarity would perform better than the general-purpose LLM.

Vision

We sought to build a lightweight, consumer-friendly product that serves as a **research guide** for people trying to self-study the law.

Users will **type in a description of their legal situation in their own words**, and we will help them find relevant Supreme Court cases that evaluated similar case facts.

In doing so, we can point users to those cases, **helping them build an intuition** for how the Court approached a case like theirs.

Functionality

I. Case scraping

The SCOTUS scraper tries to pull published opinions from the Supreme Court's public website. It 'walks' through each of slip-opinion index page for recent court terms, identifies the docket/date/case name rows, and downloads every linked PDF. For each document, it captures the case name, docket, and date, and populates SQLite with the cleaned case facts, decision summary, and any serialized related cases, honoring optional per-index or per-year limits.

II. Section segmenting

Once a PDF is downloaded, the parser jumps to the first "Syllabus" page, reads forward until it finds the italicized "Held" heading to get the case facts text, and uses that boundary to read until "delivered the opinion of the Court." to get the decision text. Any remaining opinion text after that phrase is forwarded to `case_mention_extractor()`, which identifies and stores related case citations mentioned later in the opinion.

III. Citation extraction

The case mention extraction system uses regex pattern matching to identify legal case citations in the "Name v. Name" format from Supreme Court opinion text. The extractor captures plaintiff and defendant names with specific rules for capitalization and connector words like "of" and "the", while filtering out common legal false positives such as "Id.", "See", and "E.g." as well as self-citations. The scraper integrates this extraction by parsing PDF opinions and identifying the "Held:" section, then extracting citations from either the text after "delivered the opinion of the Court" for regular opinions or from the full text after "Held:" for per curiam decisions. This dual approach ensures comprehensive citation extraction across different opinion formats, with the extracted case names stored as JSON arrays in the database and displayed in the frontend for each similar case result.

IV. Semantic Embedding

We used three different embedding models in order to capture the semantic meaning of the different sections of the Supreme Court case text. We were curious how models trained on different corpora could yield different embedding spaces and, therefore, different recommendations for similarity in the legal domain.

Selected Models:

SBERT/SentenceTransformer (all-MiniLM-L6-v2)

1. This model was initialized with nreimers' *MiniLM-L6-H384-uncased* model.
2. It was fine-tuned on 1.1 billion sentences, including:
 - a. 726 million Reddit comments, 115 million S2ORC abstracts, 77 million WikiAnswers, as well as text from Stack Exchange, Yahoo Answers, Flickr, and more.
3. It outputs a 384-dimensional embedding vector.
4. See <https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2>

nlpaueb's LEGAL-BERT-BASE (legal-bert-base-uncased)

1. This model was initialized with the base Google BERT model (*bert-base-uncased*).
2. It was then trained for 1 million steps using English-language legal text, including:
 - a. 116k docs of EU legislation, 61k docs of UK legislation, 19k European Court of Justice cases, 12k cases from the European Court of Human Rights, 164k cases from US courts (via case.law), and 76k US contracts from the US Securities & Exchange Commission
3. It outputs a 768-dimensional embedding vector.
4. See <https://aclanthology.org/2020.findings-emnlp.261.pdf> & <https://huggingface.co/nlpaueb/legal-bert-base-uncased>

Casehold's Legal-BERT (legalbert)

1. This model was also initialized with the base Google BERT model.
2. This model was then trained for 1 million steps on the entire Harvard Law case corpus
 - a. 3.4 million legal decisions across all U.S. federal and state courts from 1964-2021
3. Unlike model #2 (nlpaueb's), however, this model also applied special tokenization and segment segmentation adapted for legal text.
4. It outputs a 768-dimensional embedding vector.
5. See <https://arxiv.org/abs/2104.08671> & <https://huggingface.co/casehold/legalbert>

Observations

- SBERT is trained primarily on Reddit.
- nlpaueb's LEGAL-BERT is trained on policy and case law from Europe and America.

- Casehold’s Legal-BERT is extensively & exclusively trained on American case law.

Hypotheses

Hypothesis 1:

These three models will act quite differently, embedding text in unique ways and therefore making similarity judgments that are distinct from how the other models do.

Hypothesis 2:

We suspect that the legal BERTs will be more similar to each other than to SBERT.

Hypothesis 3:

We anticipate that one of the Legal-BERTs will be the most accurate; however, we can’t write off SBERT.

Since SBERT outputs only 384-dimensional vectors whereas the Legal-BERTs produce 768-dimensional vectors, we suspect that the Legal-BERTs will have the capacity to capture more nuance and detail from the text it processes. We suspect that this will help with accuracy.

However, since normal people will be typing up their legal situation in plain language, this will be very out-of-distribution for Casehold’s Legal-BERT model, which is exclusively fine-tuned on highly intellectual writing (official opinions as published from U.S. courts). Therefore, it may actually struggle to embed the ordinary plain language into the legalese embedding space. Conversely, official legalese will be very out-of-distribution for the Reddit-trained SBERT, so even if it properly captures the meaning contained in the user’s prompt, it will likely struggle to find similar legal cases.

V. Cosine Similarity-based Search

We used these three models to process the written case facts from the Supreme Court opinions into embedding vectors. To find the cases that were “most similar” to the provided scenario, we use cosine similarity to find the nearest embeddings. This evaluates the angle between the vectors. This rests on the premise that nearby locations in the embedding space correspond with semantically similar text. However, these models do not perfectly capture and enumerate facts, and can be distorted by tone and phrasing. Nevertheless, they provide a decent tool for evaluating similarity.

VI. User Interface

The frontend UI allows users to input case facts or legal descriptions and select from one of three models (Legal-BERT, Harvard LegalBERT, or Sentence Transformer). After they click “Find Similar Cases”, it will run inference to embed the provided text and then search for similar cases in the database using cosine-similarity with each one.

Results are displayed as cards showing each similar case’s name, similarity score percentage, and three expandable sections containing the case facts, judgment, and a list of related cases cited in the opinion. If a user clicks on Case Facts or Judgment, a section will expand to show them the full text that was extracted from that particular part of the Opinion. If they click on Case Mentions, it will list out all the other court cases that this Opinion cited. This allows the user to branch off into further reading, tracing back precedent to understand foundational legal principles that might similarly be invoked in their legal scenario.

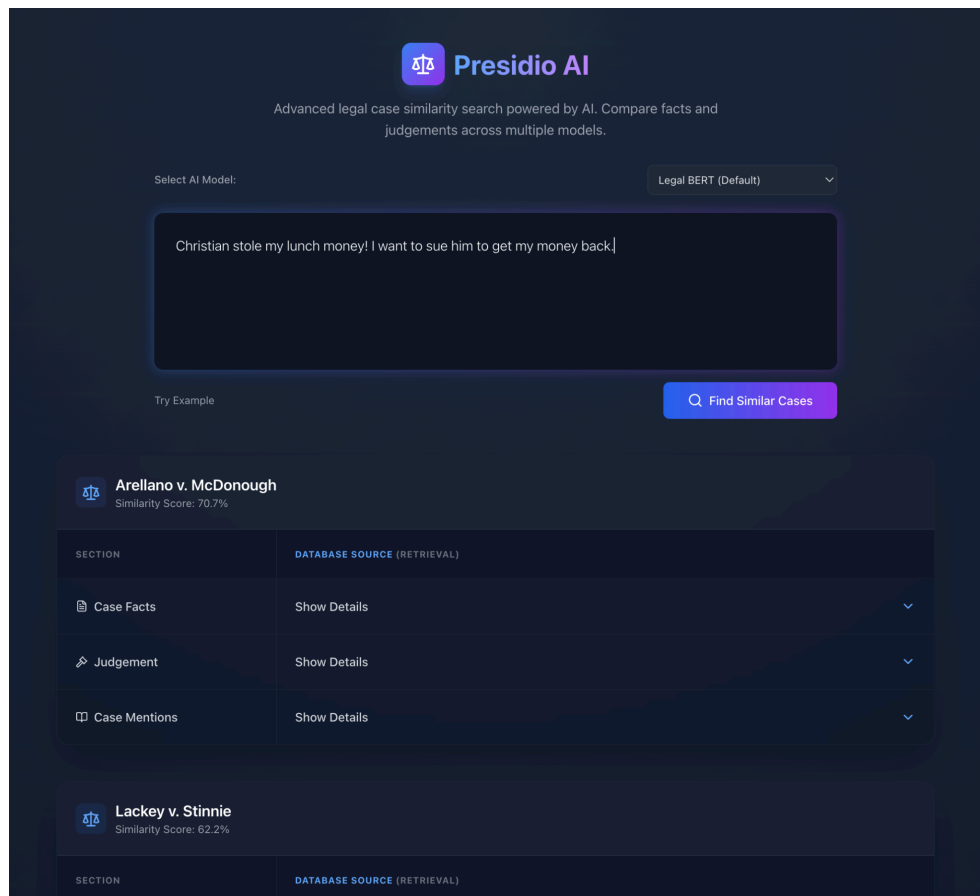


Figure 1: A snapshot of the web app.

Codebase: <https://github.com/banyasp/presidioAI>

Experiments

***Experiment 1* | “Christian stole my lunch money! I want to sue him to get my money back.”**

1. SBERT recommends:

- a. *E.M.D. Sales, Inc. v. Carrera* (similarity score = .734)
 - i. Dispute over whether workers qualified for the FLSA overtime/min-wage exemption.
- b. *Little Sisters of the Poor Saints Peter & Paul Home v. PA* (.726)
 - i. Challenge to ACA rules letting employers opt out of contraceptive coverage.
- c. *Snyder v. United States* (.724)
 - i. Mayor took bribes tied to a city trash-truck contract.

2. nlpaueb’s LegalBERT recommends:

- a. *Arellano v. McDonough* (.707)
 - i. Veteran denied retroactive VA benefits beyond one-year filing deadline.
- b. *Lackey v. Stinnie* (.622)
 - i. Virginia drivers challenged automatic license suspensions for unpaid court fines and then sought attorney’s fees after the law was repealed
- c. *Sharp v. Murphy* (.617)
 - i. Murder case turning on whether crime occurred on Creek reservation land.

3. Casehold’s LegalBERT recommends:

- a. *Arellano v. McDonough* (.963)
 - i. Retroactive VA disability benefits for a Navy veteran who filed decades after discharge.
- b. *Tandon v. Newsom* (.931)
 - i. Home religious gatherings restricted more than similar secular activities during COVID.
- c. *Dept. of Education v. California* (.929)

- i. States fight cancellation of teacher-training grants tied to DEI-focused programs.

***Experiment 2* | “I accidentally let the police into my house. They didn’t have a warrant, but they’re using what they saw in my house against me in court!”**

- a. SBERT recommends:
 - i. *Campos-Chaves v. Garland* (similarity score = .839)
 - 1. Noncitizens seek to undo in-absentia removal orders after defective initial notices.
 - ii. *United States v. Zubaydah* (.831)
 - 1. Ex-CIA detainee’s bid for contractor testimony barred by state-secrets privilege.
 - iii. *FBI v. Fazaga* (.831)
 - 1. Muslims challenge mosque surveillance; Court says FISA doesn’t displace state-secrets privilege.
- b. nlpaueb’s LegalBERT recommends:
 - i. *Tandon v. Newsom* (similarity score = .944)
 - 1. Home religious gatherings restricted more than similar secular activities during COVID.
 - ii. *Dept. of Education v. California* (.925)
 - 1. States fight cancellation of teacher-training grants tied to DEI-focused programs.
 - iii. *Ramos v. Louisiana Revisions* : 8/28/25 (.799)
 - 1. Court holds state felony convictions require unanimous juries.
- c. Casehold’s LegalBERT recommends:
 - i. *Arellano v. McDonough* (similarity score = .921)
 - 1. Veteran denied retroactive VA benefits beyond one-year filing deadline.
 - ii. *New York State Rifle & Pistol Assn., Inc. v. Bruen* (.908)
 - 1. Strikes down New York’s “proper cause” rule for public concealed carry.
 - iii. *United States v. Arthrex, Inc.* (.903)

1. PTAB judges’ unreviewable power fixed by giving PTO Director final say.

Analysis

In Experiment 1, we see mixed results. For SBERT, the first and third most similar cases discussed “getting [one’s] money back,” but they are about different contexts than the user’s scenario, which is about a robbery. Nlpaueb’s LegalBERT’s first two cases were directly about getting one’s money back, particularly for government benefits and fines.

The three models yielded quite different proposals, demonstrating the wide variability that is produced based on how a model trains on different corpora. This validates our *Hypothesis 1*.

Casehold’s LegalBERT agreed with nlpaueb’s LegalBERT on the #1 most similar case, with both focusing on *Arellano v. McDonough*. This provides evidence for our *Hypothesis 2*.

However, since all three models yielded 2 out of 3 recommended cases that were focused on recouping financial losses, it’s unclear to judge the LegalBERTs as better than SBERT at this time (*Hypothesis 3*).

In Experiment 2, there is again a medley of recommendations, with less obvious relevance to the scenario. None of the 9 proposals discussed warrantless home searches, though they allude to the prompt in different ways. SBERT focuses on what evidence or procedure can be used in court, honing in on the idea of the police using something “against me in court!” This can still offer helpful insight into what is permissible and impermissible evidence, which can still help the user prepare their case. Nlpaueb’s LegalBERT’s top recommendation focuses on the government intruding on activities in the home, which covers that other part of the scenario. Casehold’s LegalBERT appears to be very disjointed and unrelated.

Once again, the recommendations varied widely, providing further evidence for *Hypothesis 1*.

This time, the top 3 cases for the LegalBERTs didn't have any overlap, reducing our confidence in *Hypothesis 2*.

The quality rankings here seem to be: SBERT > nlpaueb's LegalBERT > Casehold's LegalBERT. Perhaps this is because the user's input is framed so colloquially, from the first person, with emotional exclamations. This would be well represented in SBERT's training set, somewhat in nlpaueb's scattered legislation and policy set, and almost not at all in the high legal academia writing of Casehold's LegalBERT. This undermines *Hypothesis 3*.

Taken together, our *Hypothesis 1* on different training sets yielding different outputs appears true. *Hypothesis 2* on the relative consistency between the LegalBERTs appears possibly true, but would require further study. *Hypothesis 3* on the quality of LegalBERTs exceeding SBERT appears less likely, aligning with our original nod to the match in the distributions of SBERT's training set and the ordinary user's submission.

Discussion

In this project, we developed a tool to help ordinary people get started with legal research when they have no experience. We invite them to speak plainly, and then match them with the court cases that may be relevant.

This system is imperfect, as our experiments suggest, and it's important to note that in legal scenarios, there is not just one meaning but rather a combination of multiple features. Cosine similarity, as applied to the entire text embedding, can therefore find alignment according to any of those features.

Although LegalBERTs might seem to be the optimal choice for law-related natural language processing, our experiments suggest some further reflection. Although the law is a precise endeavor, ordinary humans are not, and therefore to make a system that can 'port' into that colloquial style, it may be best to use a BERT that is used to handling human beings. You wouldn't put a sheltered, elite math nerd on the streets and expect him/her to be good at empathizing and intuiting slang and gruff speech. Casehold's LegalBERT is that way—entirely untrained in ordinary human speech, and therefore perhaps less effective at making that connection.

Therefore, our key finding is that natural language is organic, flexible, and diverse. Just as humans learn to speak in different styles with the jargon of different disciplines or cultures, so too do models. Therefore, in conducting NLP, it is essential to ensure that great care is taken to ensure that the model chosen is fit for the task.